5 **(a)** **(i)** The magnetic flux density of a magnetic field is defined as the force per unit current per unit length (of conductor) acting on a straight current-carrying conductor placed at right angles to the magnetic field [B1].

(ii) Current in clockwise direction through wire [A1]

(iii)

$$B = \mu_o nI = 4\pi \times 10^{-7} \times \frac{50}{0.30} \times 2.0 = 4.189 \times 10^{-4}\ \mathrm{T}\ [\mathrm{A1}]$$

(iv)

$$F = Bqv$$

$$= (4.189 \times 10^{-4})(1.6 \times 10^{-19})(10)\ [\mathrm{C1}]$$

$$= 6.7 \times 10^{-22}\ \mathrm{N}\ [\mathrm{A1}]$$

(v) The magnetic force is acting perpendicular to the direction of travel/ velocity of the electron [M1] and this will only affect direction of velocity and not its magnitude [A1]

Alternative answer:

The magnetic force is acting perpendicular to travel/ velocity of the electron [M1] and therefore no (net) work done [A1] on the electron, resulting in no change in kinetic energy

(b) **(i)** No net force acting on electron,

Magnetic force= Electric force [C1]

qE= Bqv

E= Bv = 4.19 x 10^{-3} Vm^{-1} [A1]

(ii) Downwards vertical arrow labelled E [B1]

6 **(a)** α : protons decrease by 2, neutron decrease by 2 [B1]